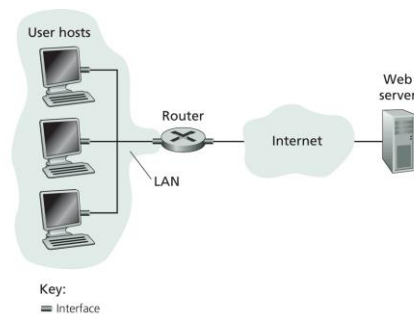


03 Medium Access Control Sublayer

03.02 Ethernet

LAN: introduction

- Multiple access protocols are used in various cases:
 - Wireless channels
 - Satellite links
 - LAN (see figure)

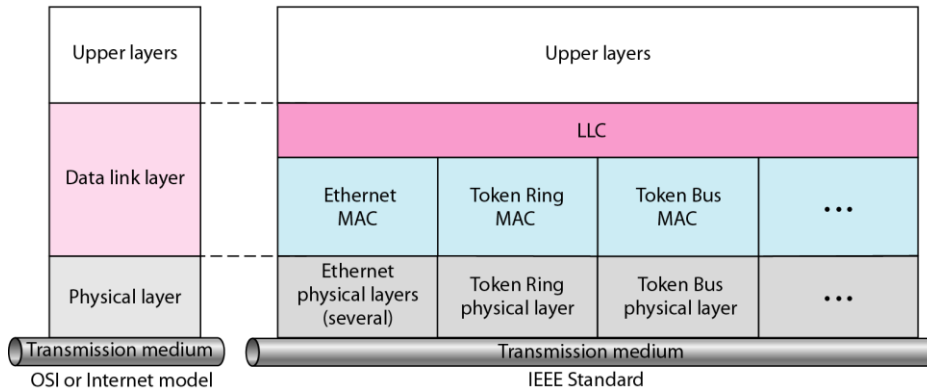


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- ♦ User hosts access an Internet Web server through a LAN. The broadcast channel between a user host and the router consists of one link.

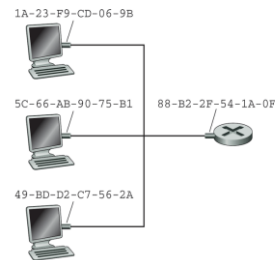
IEEE standards for LAN

LLC: Logical link control
MAC: Media access control



LAN: MAC addresses

- Nodes in LANs have data link addresses (we will see that they also have network level addresses (i.e. IP addresses))
- Data link address is referred to NIC and is known as MAC (Medium Access Control) address
- In many LANs (Ethernet and WLAN 802.11) MAC addresses are 6 byte long $\rightarrow 2^{48}$ possible addresses
- MAC address of each adapter is permanently embedded in its ROM when constructed
- There are no 2 equal MAC addresses (IEEE supervises the attribution)
- Destination MAC address is put in the packets that need to be sent on a LAN
- The right destination node recognizes packets sent to it.
- Broadcast addresses are possible $\rightarrow$ FF-FF-FF-FF-FF-FF

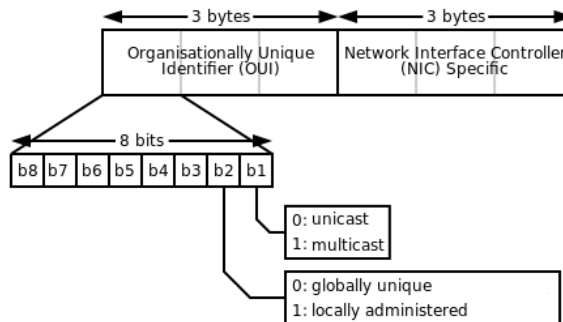


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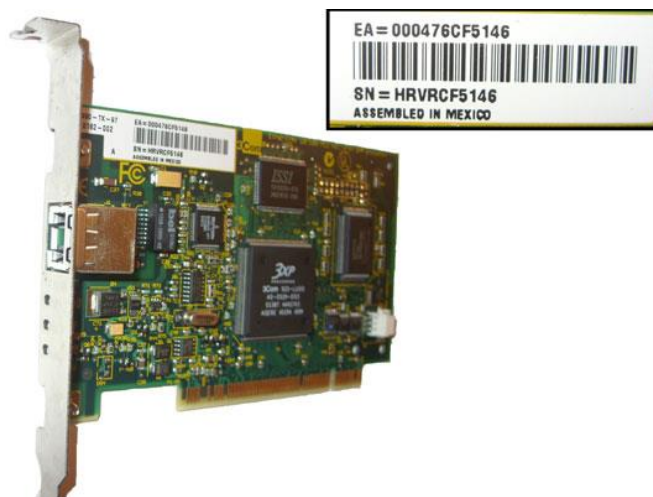
♦ Each adapter connected to a LAN has a unique MAC address.

LAN: MAC addresses

- Broadcast addresses are possible → FF-FF-FF-FF-FF-FF
- Less significative bit of first Byte:
 - If 0 is a single address
 - Otherwise is a group address
- Administered
 - 0 → unique identified
 - 1 → LAN Admin can administrate MAC Address



MAC Address and NIC

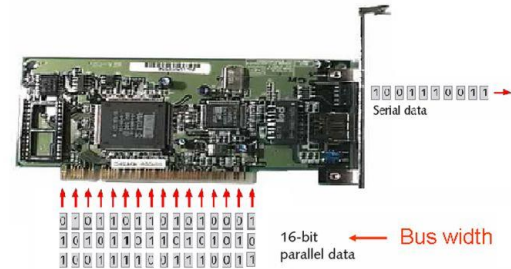


MAC Addresses and NIC

Network Interface Card (NIC) and MAC address



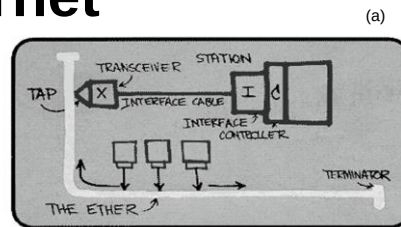
NIC Transmission: From Parallel to Serial and Vice Versa



NICs mediate communication between a computer and the network cable

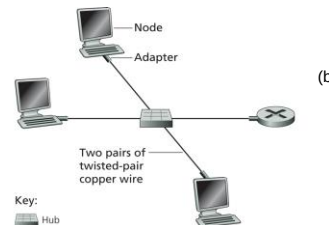
Ethernet

- It was born in '70s
- Today it is for LAN what Internet is for global internetworking.
- Why?
 - First high speed widely diffused
 - Ever growing speed
 - Switched Ethernet proposed in '90s
 - Diffusion has led to economies of scale
- Now newer networks have a star configuration (see (b))



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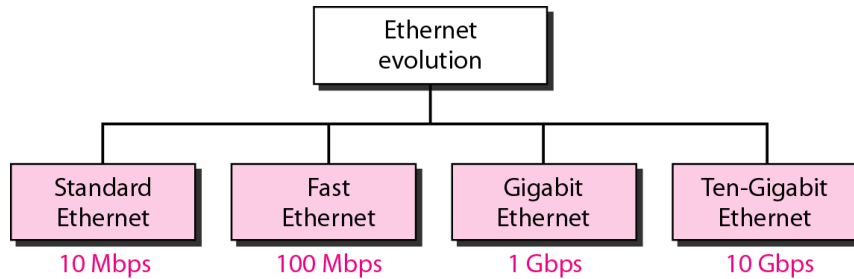
- ♦ The original Metcalfe design led to the 10Base5 Ethernet standard, which included an interface cable that connected the Ethernet adapter to an external transceiver.



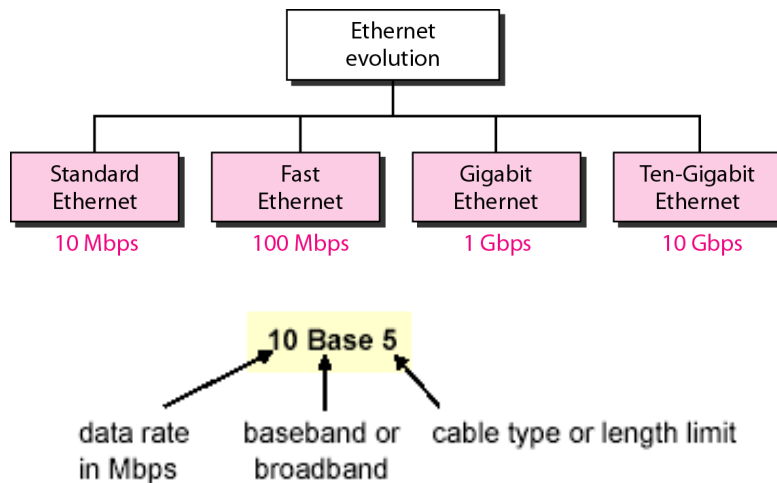
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- ♦ Star topology for Ethernet. Nodes are interconnected with a hub.

Ethernet evolution: 4 generations



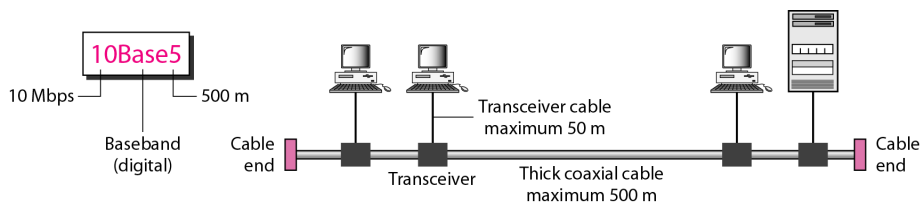
Traditional Ethernet Categories



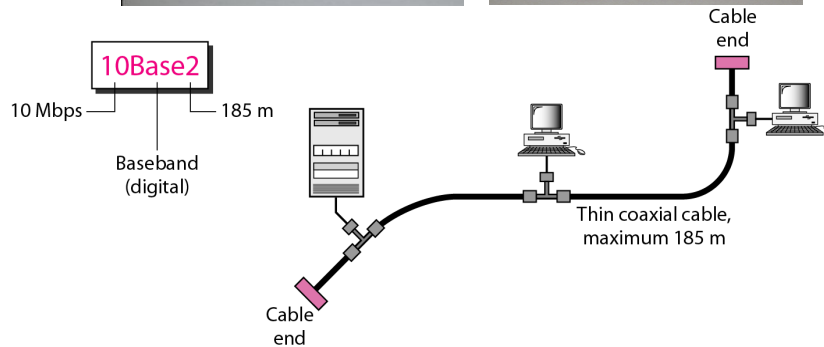
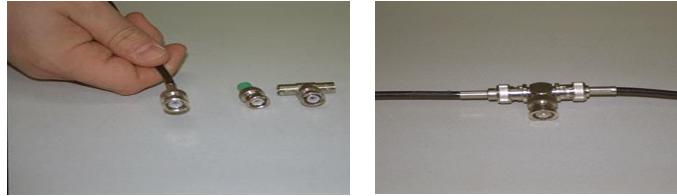
IEEE 802.3 Cable Types

Name	Cable Max.	Max Cable Segment Length	Nodes /segment	Topology
10Base5	thick coax	500 meters	100	Bus
10Base2	thin coax	185 meters	30	
10BaseT	twisted pair	100 meters	1	Star
10BaseF	Fiber Optic	2Km	1	Star

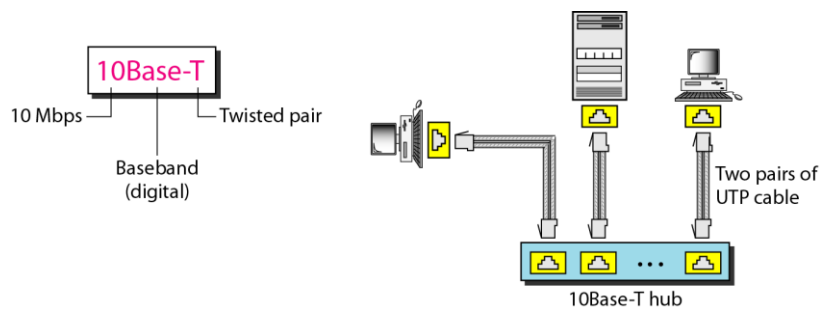
Ethernet: 10Base5



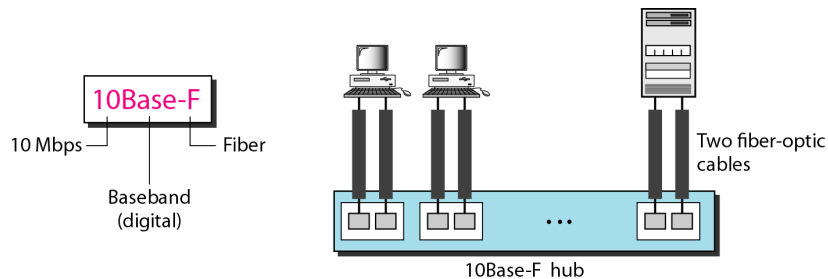
Ethernet: 10Base2



Ethernet 10BaseT



Ethernet 10BaseF

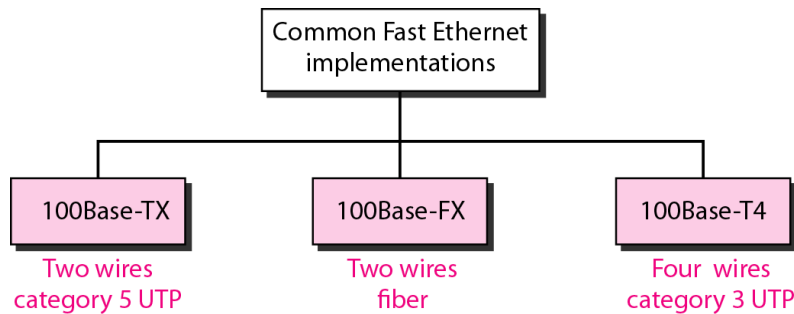


Fast Ethernet

- 100 Mbps transmission rate
- same frame format, media access, and collision detection rules as 10 Mbps Ethernet
- can **combine 10 Mbps** Ethernet and Fast Ethernet on same network using a **switch**
- media: twisted pair (CAT 5) or fiber optic cable (no coax)
- Star-wire topology
 - Similar to 10BASE-T

Name	Cable	Max. segment	
100Base-T4	Twisted pair	100 m	CAT 3
100Base-TX	Twisted pair	100 m	CAT 5
100Base-FX	Fiber optics	2000 m	

Fast Ethernet Implementation



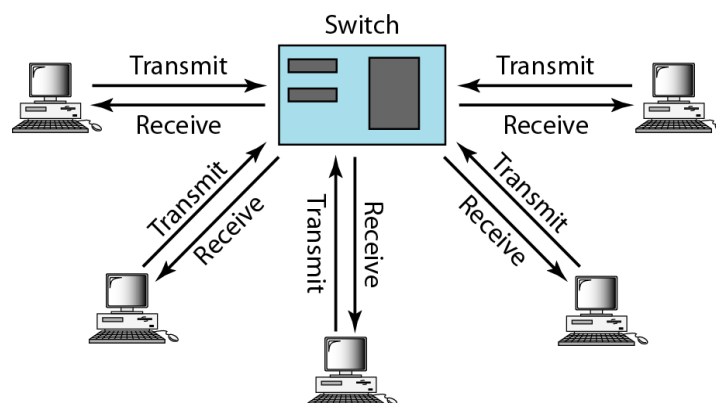
Fast Ethernet Operation

- Traditional Ethernet is half duplex
 - Either transmit or receive but not both simultaneously
- With full-duplex, station can transmit and receive *data* simultaneously
- With full duplex, Throughput (actual transmission rate) is doubled.
 - 10-Mbps Ethernet in full-duplex mode, theoretical transfer rate becomes 20 Mbps
 - 100-Mbps Ethernet in full-duplex mode, theoretical transfer rate becomes 200 Mbps

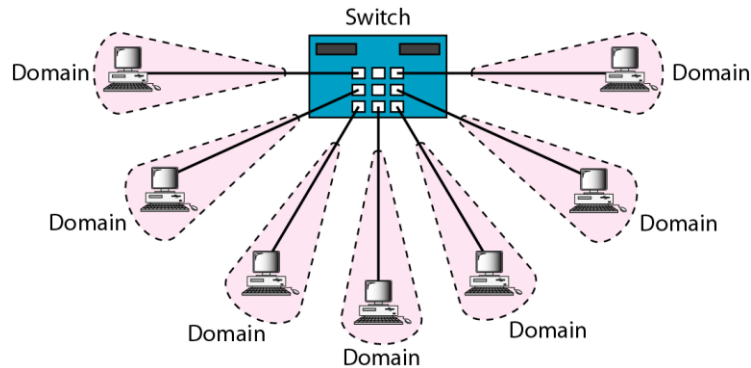
Fast Ethernet Operation

- Changes that should be made with any computer in order to operate in Full-Duplex Mode
 - Attached stations must have full-duplex NIC cards
 - Must use two pairs of wire one pair for transmitting from host to switch (inbound) and the other pair for transmitting from switch to host (outbound)
 - Must use a switch as a central device not a hub
 - Devices must be connected point-to-point (dedicated) to the switch
 - Each station constitutes separate collision domain
 - CSMA/CD algorithm no longer needed (no collision)
 - No limit on the segment length
 - Same 802.3 MAC frame format used

Full Duplex Switched Ethernet



Switched Ethernet



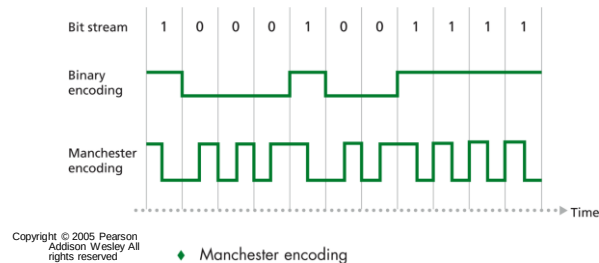
Ethernet packet structure

- **Data:** (46-1500 byte). MTU (Maximum Transfer Unit) = 1500 byte → network layer, in case, must fragment its packets or stuff them with dummy bytes
- **Destination address:** (6 byte). Packets with other address are discarded
- **Source address:** (6 byte). The MAC address of source
- **Type:** (2 byte). Protocol (de)multiplexing
- **CRC:** (4 byte). CRC is computed by transmitter as a function of the bits in packet (excluded the preamble). Receiver checks the CRC.
- **Preamble:** (8 byte). First 7 byte = '10101010', last byte = '10101011'. Can be used by receiver to synchronize with transmitter clock.



Ethernet

- Ethernet uses base band transmission
- Many Ethernet (e.g. 10BaseT) use *Manchester encoding* to allow synchronization since every bit is associated to a transition



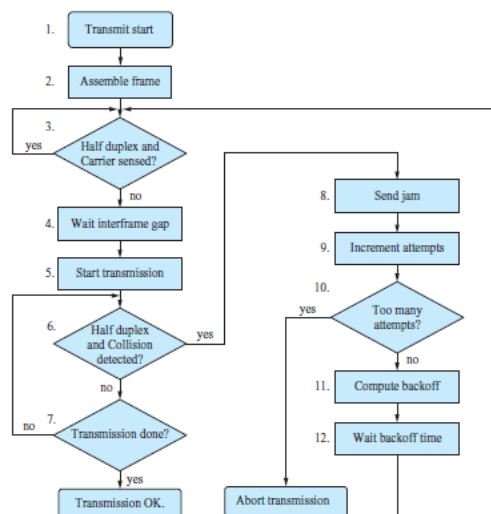
Ethernet: CSMA/CD

- When nodes of the LAN are connected by a hub, if one node sends a packet, all nodes receive it → need multiple access protocol
- Ethernet uses CSMA/CD:
 - no slot
 - carrier sense
 - Collision detection
 - Re-transmission after a short time
- Good efficiency (around 100% with bounded delay)

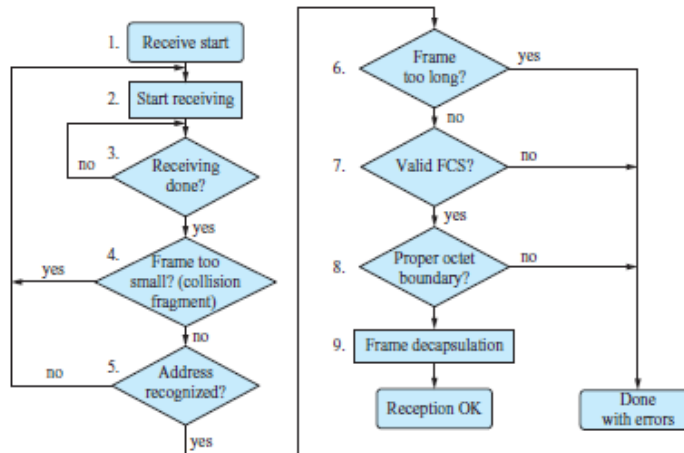
Ethernet: CSMA/CD

- Adaptor operation steps:
 - Take a network-level datagram, prepare Ethernet packet, put it in buffer
 - If channel inactive (free for >96 bit of time) start transmission, otherwise wait until free
 - During transmission check for signals from other adapters
 - If detects signals from other adapters, stop transmission and send a 48 bit noise signal (jam)
 - Wait for k -times the time for sending 512 bit and then come back to second step, where k is computed as:
 - k = casual value in $[0, 2^m - 1]$ and $m = \min\{n, 10\}$, and n is the number of consecutive collision detected
 - This algorithm is known as **exponential backoff** since the cardinality of the set in which k is chosen grows exponentially (up to $n=10$)

Frame transmission flow CSMA/CD



Frame reception flow CSMA/CD



Ethernet: CSMA/CD

- Exponential backoff algorithm is adaptive
- Efficiency: average fraction of time during which packets are on channel without collisions (with large number of hosts). It can be approximately evaluated as:

$$Eff = \frac{1}{1 + 5t_{prop}/t_{trasm}}$$

- Where:

- t_{prop} = max propagation time between 2 nodes
- t_{trasm} = time to send MTU (around 1.2 ms for 1 Mbps Ethernet)

Gigabit Ethernet

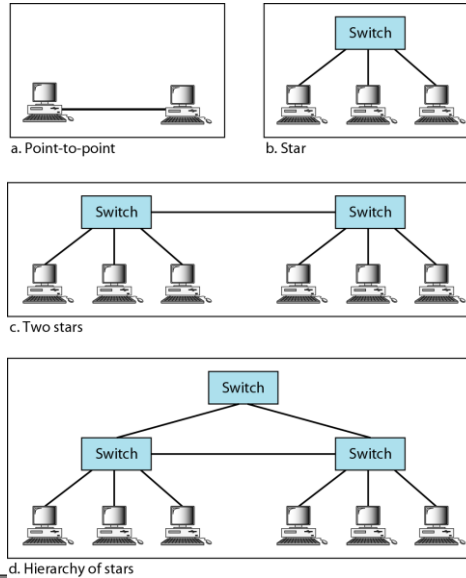
- Speed 1Gbps
- Minimum frame length is 512 bytes

Name	Cable	Max. segment	Advantages
1000Base-SX	Fiber optics	550 m	Multimode fiber (50, 62.5 microns)
1000Base-LX	Fiber optics	5000 m	Single (10 μ) or multimode (50, 62.5 μ)
1000Base-CX	2 Pairs of STP	25 m	Shielded twisted pair
1000Base-T	4 Pairs of UTP	100 m	Standard category 5 UTP

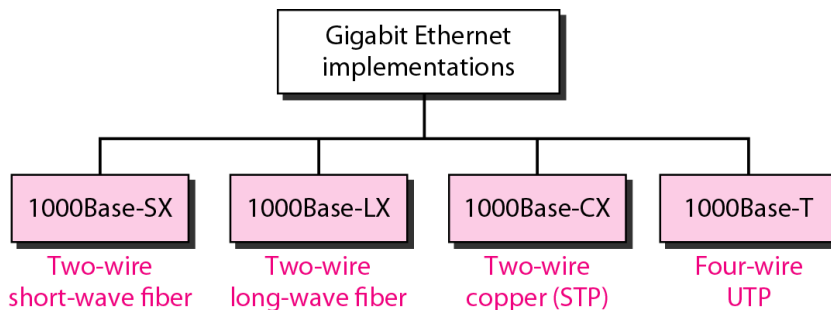
Gigabit Ethernet

- In the full-duplex mode of Gigabit Ethernet, there is no collision;
- the maximum length of the cable is determined by the signal attenuation in the cable
- Gigabit Ethernet can be shared (hub) or switched.
- Shared Hub
 - Half duplex: CSMA/CD **with MAC changes:**
 - Carrier Extension
 - Frame Bursting
- Switch
 - Full duplex: Buffered repeater *called* **{Buffered Distributor}**

Gigabit Ethernet Topologies

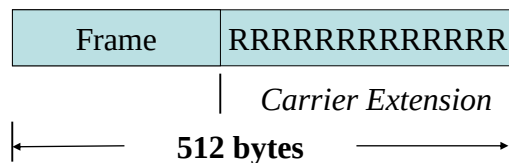


Gigabit Ethernet Implementations

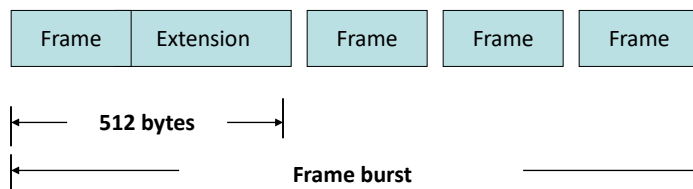


Carrier Extension

- For **10BaseT** : 2.5 km max; **slot time = 64 bytes**
- For **1000BaseT**: 200 m max; **slot time = 512 bytes**
- **Carrier Extension** :: continue transmitting control characters [R] to fill collision interval.
- This permits minimum 64-byte frame to be handled.
- Control characters discarded at destination.
- *For small frames net throughput is only slightly better than Fast Ethernet.*

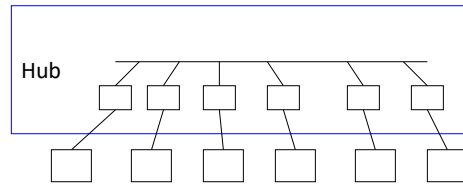


Frame Bursting



- Source sends out burst of frames without relinquishing control of the network.
- Uses Ethernet Interframe gap filled with extension bits (96 bits)
- Maximum frame burst is 8192 bytes
- *Three times more throughput for small frames.*

Buffered Distributor



- A **buffered distributor** is a new type of 802.3 hub where incoming frames are buffered in FIFOs.
- CSMA/CD arbitration is inside the distributor to transfer frames from an incoming FIFO to all outgoing FIFOs.
- 802.3x frame-based flow control is used to handle congestion.
- All links are full-duplex.

10Gbps Ethernet

- Maximum link distances cover 300 m to 40 km
- Full-duplex mode only
- No CSMA/CD
- Uses optical fiber only